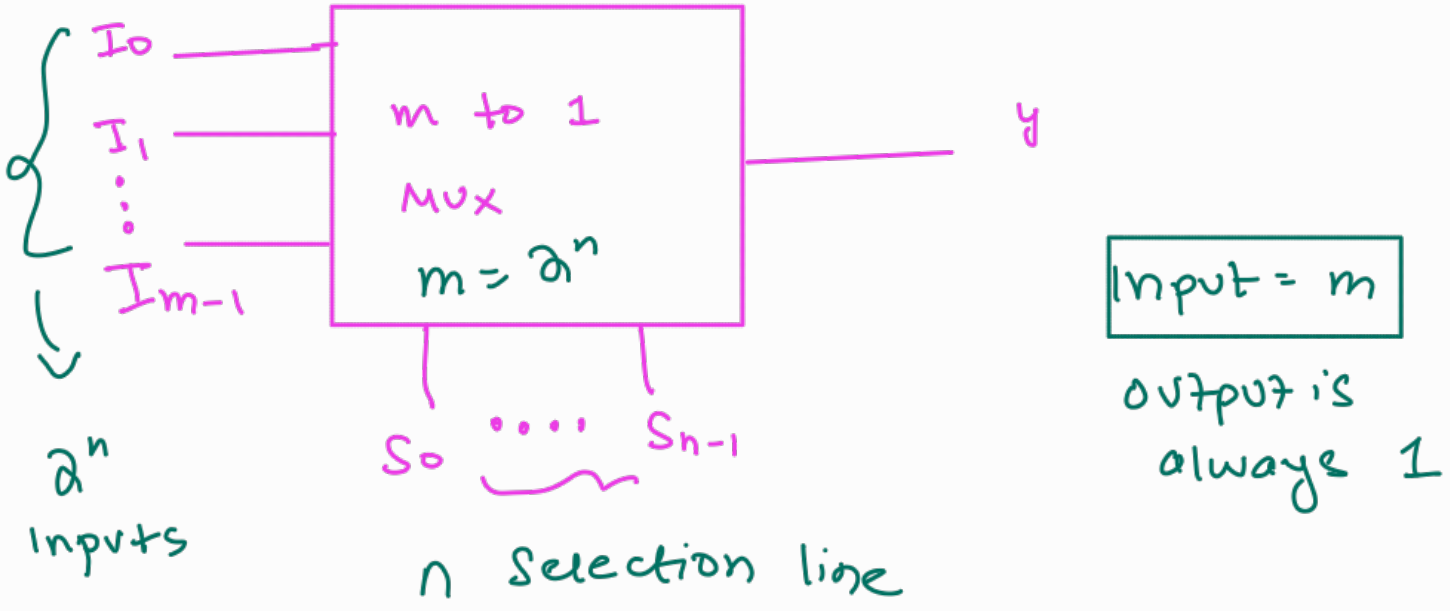
Mux: A combinational circuit that selects binary info from one of many input lines and directs to a single output line.

Mux a.ka Data Selector. It is useful for choosing from sets of data.



Selection input selects which input line to go at the output

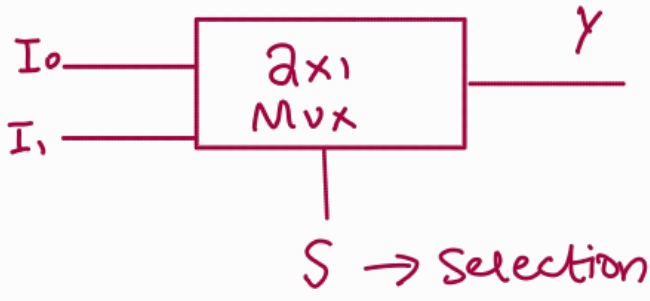
To decide total no-of selection lines we use formula $2^n = \text{inputs}$ where n is the no-of selection lines

2 to 1 Mux 2 → Inputs 1 → Outputs

To decide total no of Selection lines.

$2 = 2^n$ ⇒ n=1 Selection line

S	I_0	I_1	Y
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	1



2^3 = 8 combinations

when S=0 then I_0 value is copied exactly.

when S = 1 then I_1 value is copied exactly in the output.

convert Selection bit to decimal equivalent → I_d.

S	y
0	I_0
1	I_1

Eq condensed T.T

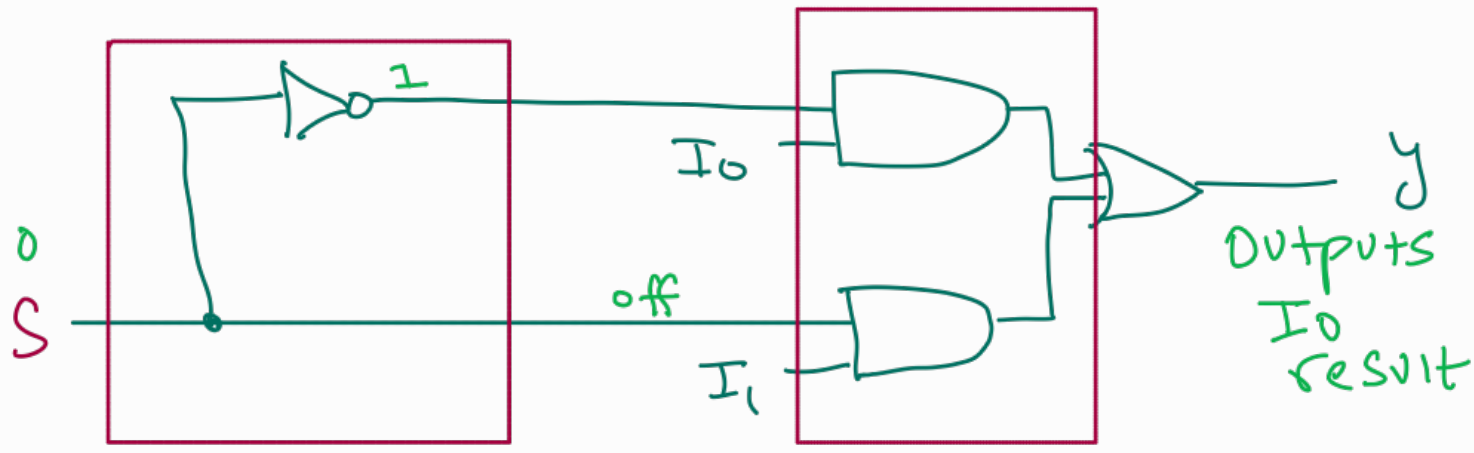
output can be I_0 or I_1 hence '+'

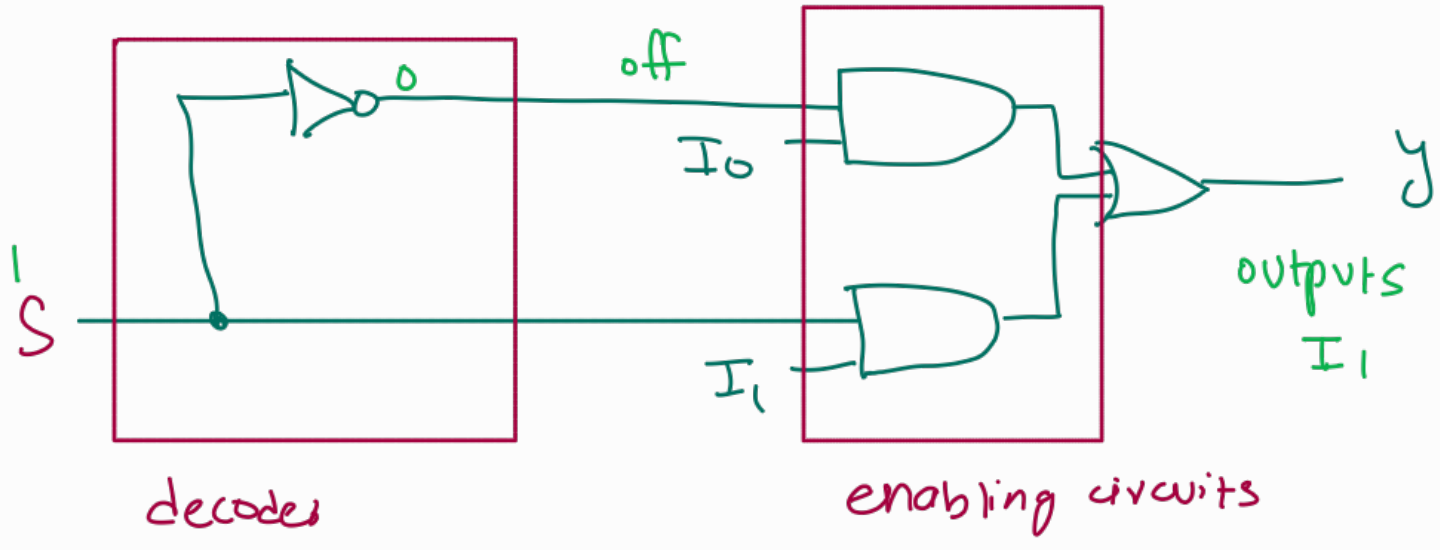
$$y = \bar{S}_0 I_0 + S_0 I_1$$

↓

SOP

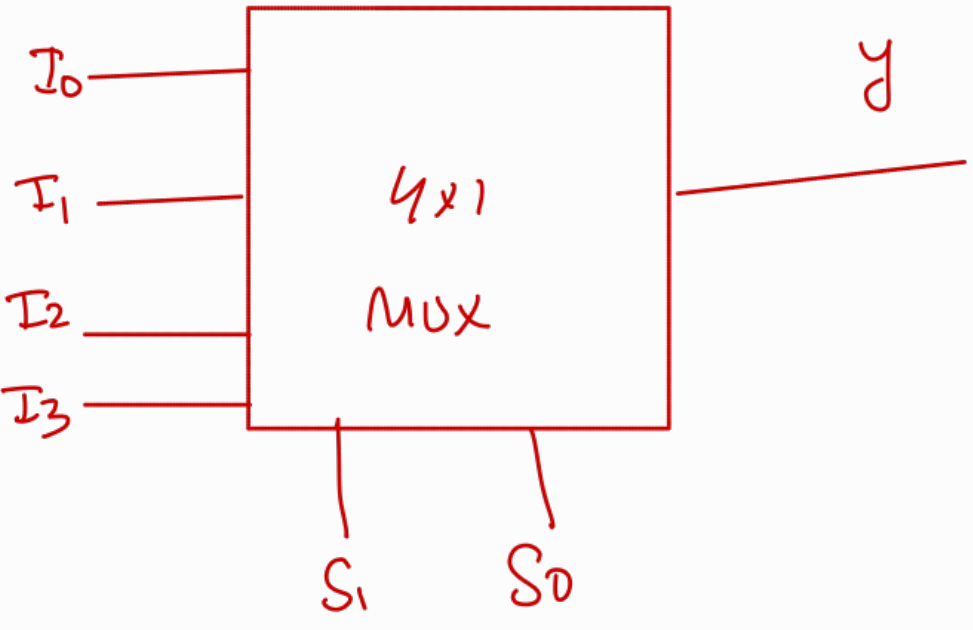
Circuit diagram





4x1 MUX total selection lines

$2^n = 4 \quad n = 2$



condensed T·T

Always make a condensed T·T

S_1	S_0	y
0	0	I_0
0	1	I_1
1	0	I_2
1	1	I_3

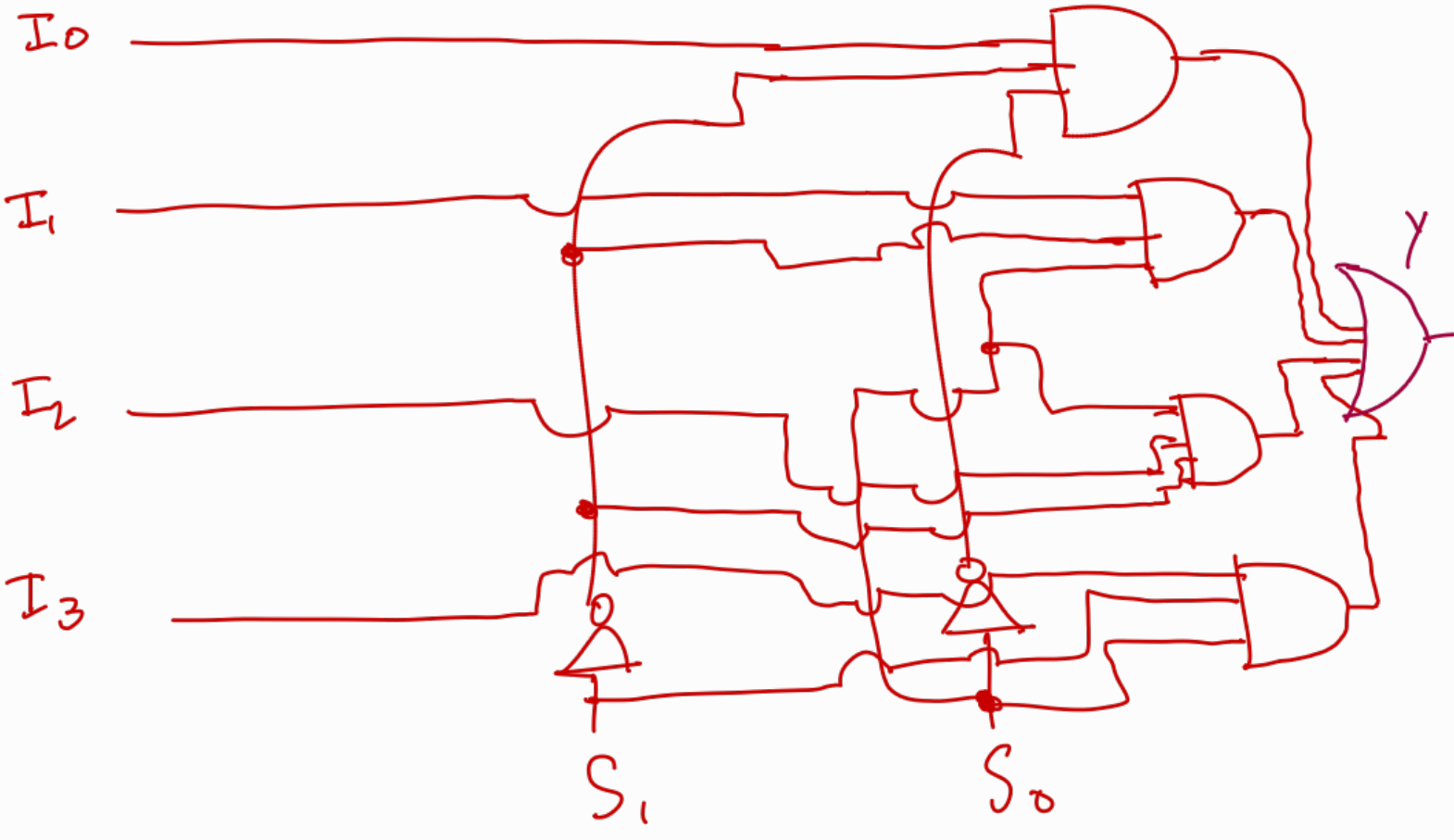
→ decimal equalent

2 Selection inputs

hence $2^2 = 4$ possible combinations.

Equation

$$y = \overline{S_1}\,\overline{S_0} I_0 + \overline{S_1} S_0 I_1 + S_0 \overline{S_1} I_2 + S_0 S_1 I_3$$

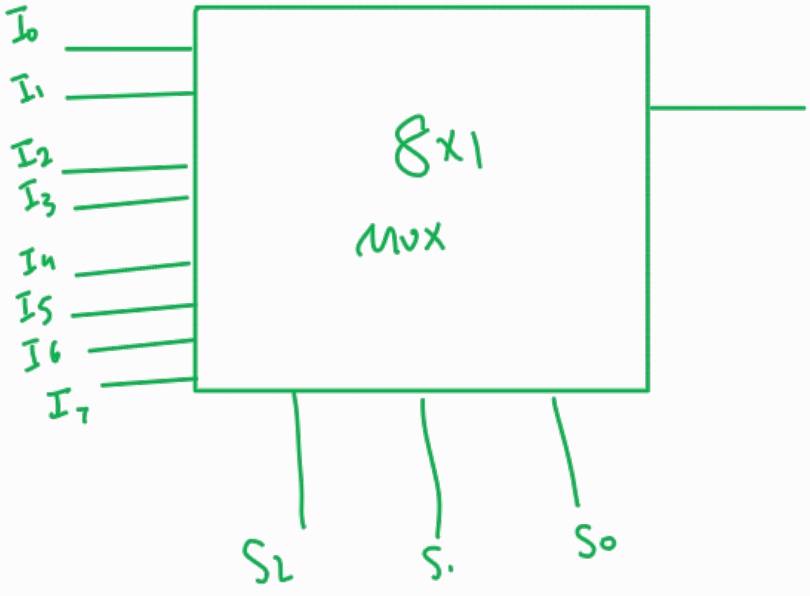


Calculate 8x1 MUX

Total Selection lines

$8 = 2^n$

$n = 3$



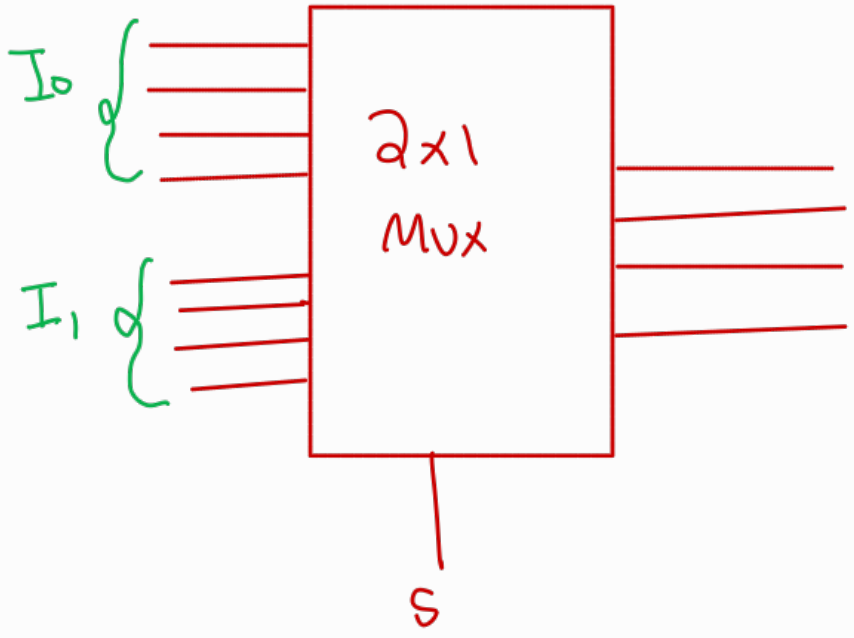
S_2	S_1	S_0	y
0	0	0	I_0
0	0	1	I_1
0	1	0	I_2
0	1	1	I_3
1	0	0	I_4
1	0	1	I_5
1	1	0	I_6
1	1	1	I_7

$$y = \overline{S_2}\,\overline{S_1}\,\overline{S_0}\,I_0 + \overline{S_2}\,\overline{S_1}\,S_0\,I_1 + \cdots\cdots$$

E.g 64 x1 MUX total no. of selection lines

$64 = 2^n = 6$

2 to 1 Quadruple MUX



Quadruple means input lines now carry 4 bits. meaning Size of I gets Say I_0 is now 4-bits

Since input of 1 number consist of 4 bits the output will also be of 4 bits.

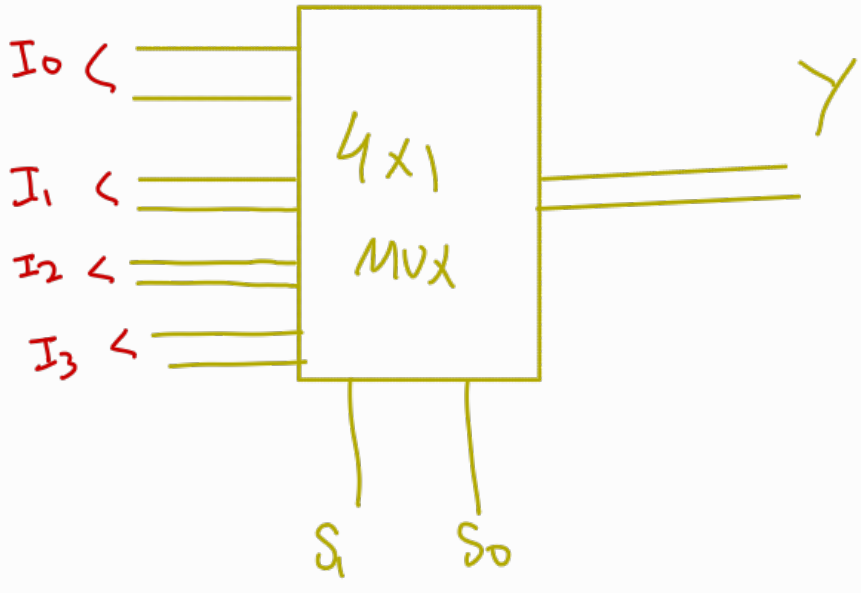
S	y
0	0000 | Select I_0
1	0001 | Select I_1

note:- Active low enable means that when E=1 the output is off. By using Inverter.

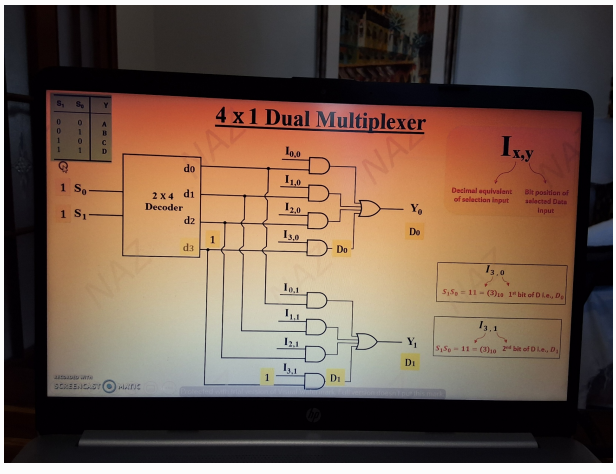
Active High means that when E=1 it gives an output. no use of Inverter.

4x1 Dual MUX

Dual means each data input lines has 2 bits

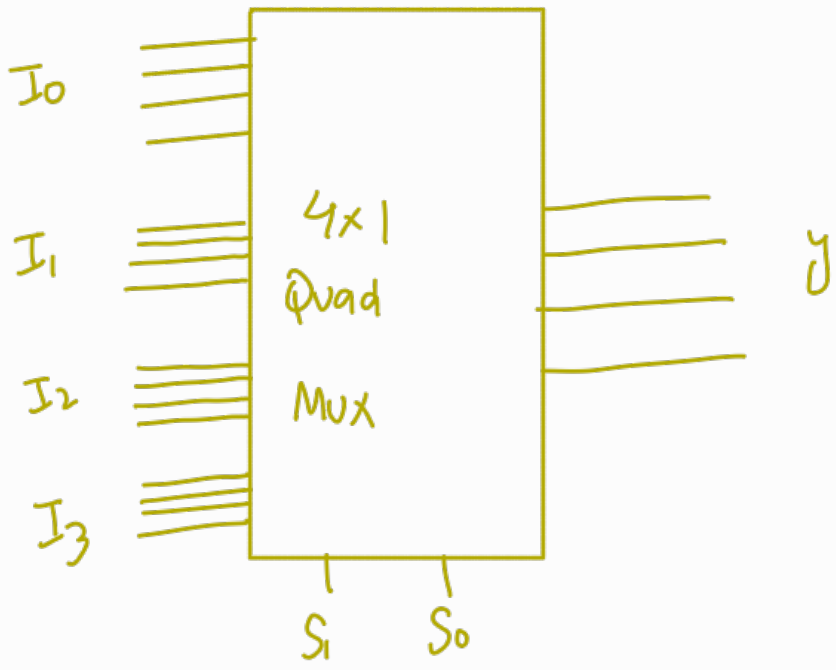


S_1	S_0	Y
0	0	I_0
0	1	I_1
1	0	I_2
1	1	I_3



4x1 Quad Mux.

S_1	S_0	y
0	0	0000
0	1	0001
1	0	0010
1	1	0011



Implementing boolean function using MUX

A boolean function of n variables can be implemented with a mux of (n-1) selection lines.

→ Boolean function is first represented in form of a T.T

→ The first n-1 variables (MSB) are applied at selection lines.

→ The remaining variable is applied as input

→ for each combination of selection lines, output is evaluated as a function of last variables.

Function can have a value of 0, 1, variables, complement of variable

Step 1: T.T of function.

Inputs			F
x	y	z	
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

F is output

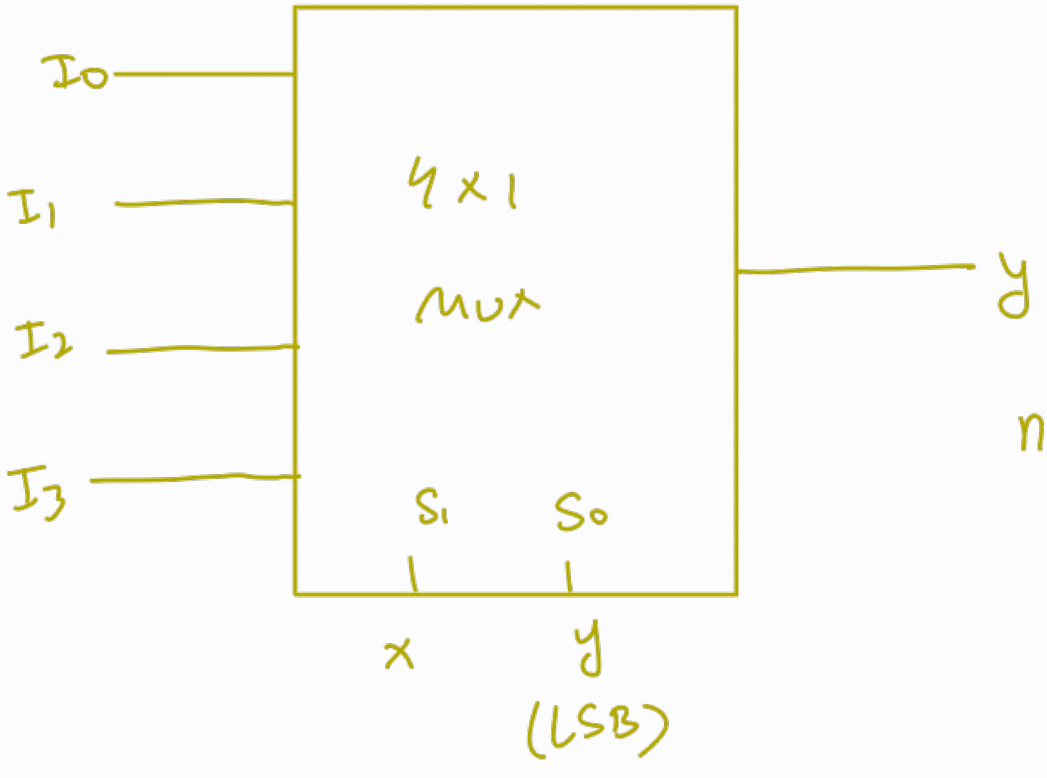
Total variables = n = 3

Selection inputs = n-1 = 3-1 = 2

For deciding Mux we use this formula:

$$2^{(n-1)} \times 1 \text{ Mux} = 2^{(2)} \times 1$$

$$= \boxed{4 \times 1}$$



note:- we always prefer selection inputs that are MSB.

Z is not counted as a selection input as it is on LSB in the t.t.

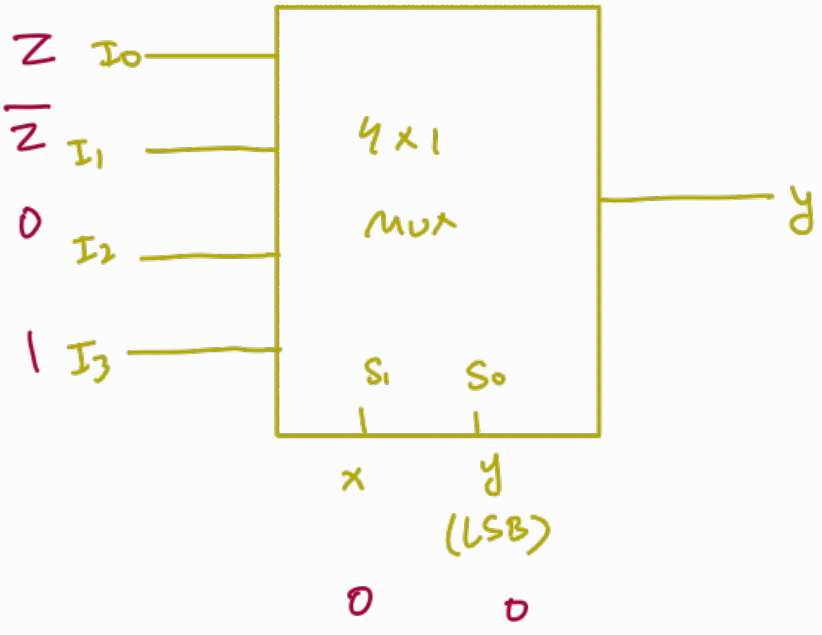
Z now will be counted as an input.

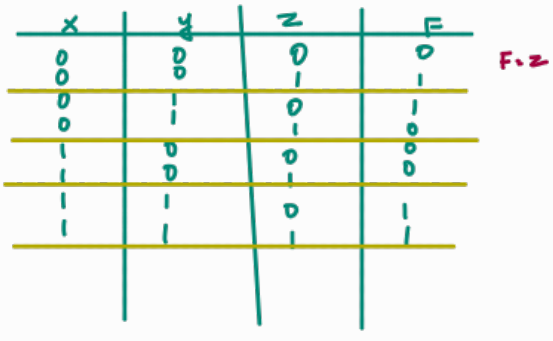
Now dividing the table into parts.

x	y	z	F
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	1

F = z

how to divide the table :- It depends on no. of selection inputs As they are two. Divide the t.t into two parts.





when x and y is 0 0 then it activities I_0 in which z is being fed into I_0

Hence I_0 takes input 0

and 1 as shown in 2 column. The relationship b/w F and Z is equal hence $\boxed{F = Z}$

X	y	Z	F	
0	0	0	0	F = Z
0	0	1	1	
0	1	0	1	F = Z' → activities I_1
0	1	1	0	
1	0	0	0	F = 0 → I_2
1	0	1	0	
1	1	0	1	F = 1 → I_3
1	1	1	1	

Example:

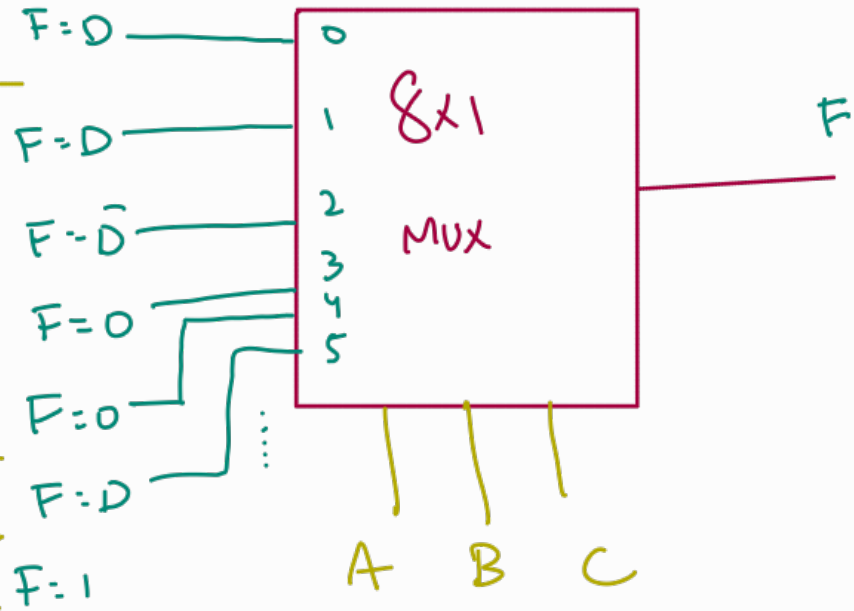
$$F(A, B, C, D) = \Sigma m(1, 3, 4, 11, 12, 13, 14, 15)$$

Total variables = 4

Total Selection lines = 4 - 1 = 3

Mux formula $(n-1) \times 1 = 2^3 \times 1 = 8 \times 1$ MUX

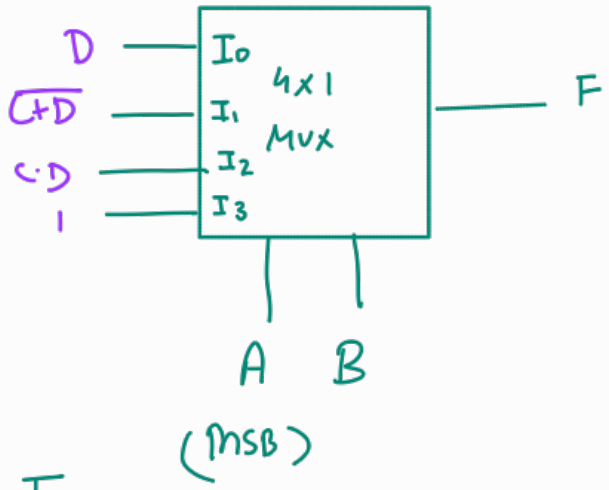
A	B	C	D	F	
0	0	0	0	0	F = D
0	0	0	1	1	
0	0	1	0	0	F = D
0	0	1	1	1	
0	1	0	0	1	F = D'
0	1	0	1	0	
0	1	1	0	0	F = 0
0	1	1	1	0	
1	0	0	0	0	F = 0
1	0	0	1	0	
1	0	1	0	0	F = D
1	0	1	1	0	
1	1	0	0	1	F = 1
1	1	0	1	1	
1	1	1	0	1	F = 1
1	1	1	1	1	



Relating D with F.

Implement $F: (A,B,C,D) = \Sigma m(1,3,4,11,12,13,14,15)$ using 4x1 MUX having A and B as Selection Inputs and C and D as data Inputs ?

A	B	C	D	F	
0	0	0	0	0	F = D
0	0	0	1	1	
0	0	1	0	0	
0	0	1	1	1	
0	1	0	0	1	$F = \overline{C+D}$ → realized through NOR.
0	1	0	1	0	
0	1	1	0	0	
0	1	1	1	0	
1	0	0	0	0	$F = C \cdot D$
1	0	0	1	0	
1	0	1	0	0	
1	0	1	1	0	
1	1	0	0	1	$F = 1$
1	1	0	1	1	
1	1	1	0	1	
1	1	1	1	1	



Relating C and D with output F.

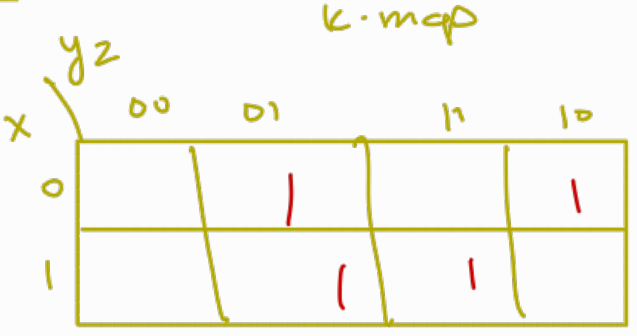
Alternative method to implement Boolean function using MUX by using k-map

Total variables = 3
Selection :- 3-1 = 2

$2^{n-1} \times 1 =$ 4x1 MUX

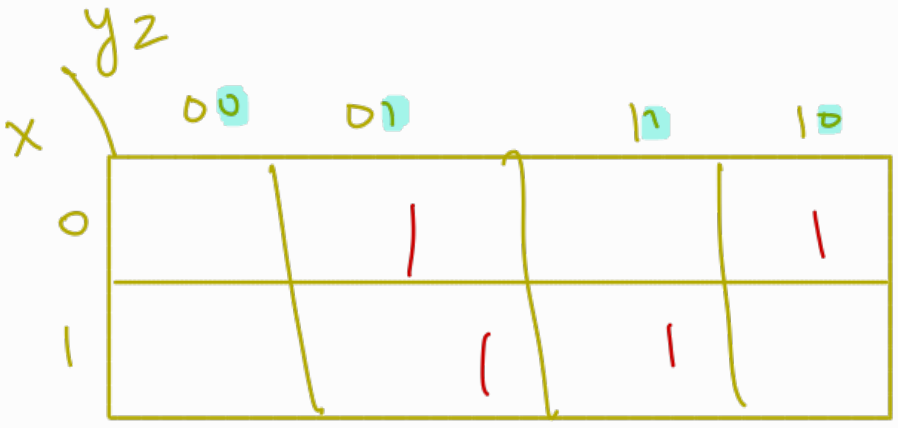
x	y	z	F
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	1

3-var k-map



Selection input is x and y.

and Z is input value → Just map the one's as we normally do.



difficult method ^ !!

Mux Implementation of 1 bit Adder

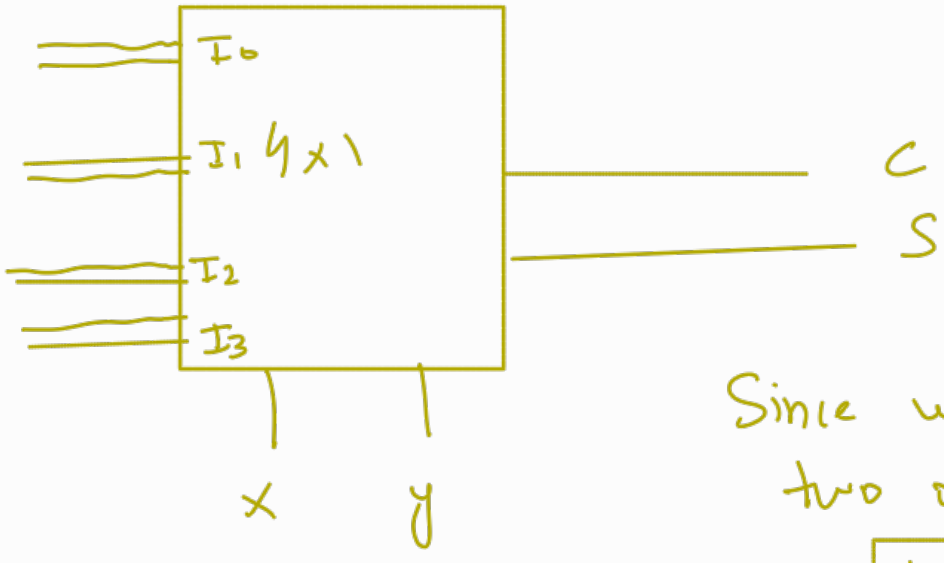
x	y	z	C	S
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	0	1
1	0	1	1	0
1	1	0	1	0
1	1	1	1	1

2 output bits / Carry and Sum :. Total variables = 3

Selection = 2

Mux Size $2^{(n-1)} \times 1$

$2^2 \times 1$ => 4x1



Since we have two outputs we use 4x1 dual MUX.

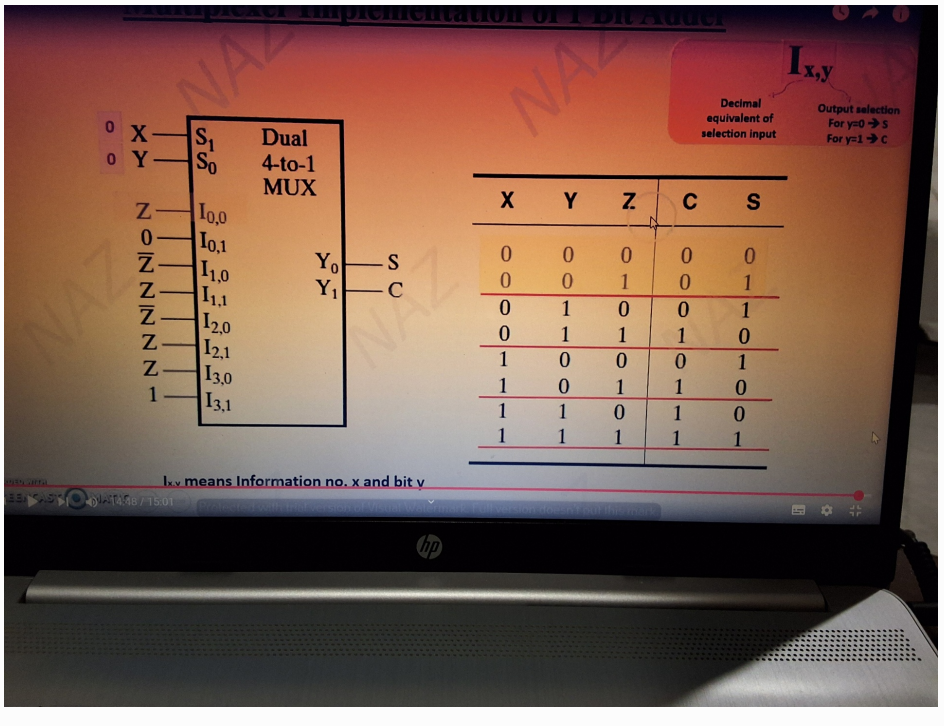
x	y	z	C	S
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	0	1
1	0	1	1	0
1	1	0	1	0
1	1	1	1	1

$S = z$
$C = 0$

$S = \bar{z}$
$C = z$

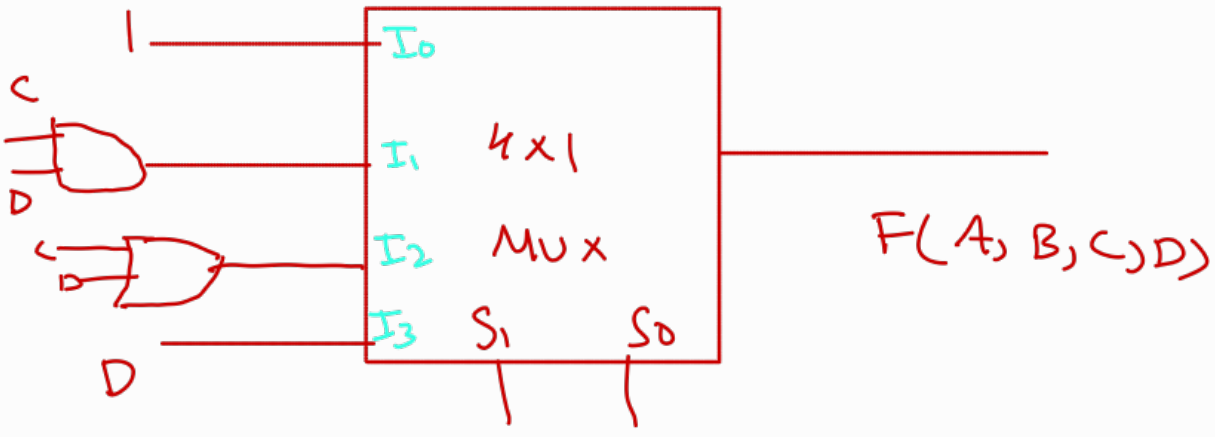
$S = \bar{z}$
$C = z$

$S = \bar{z}$
$C = 1$



X	Y	Z	C	S
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	0	1
1	0	1	1	0
1	1	0	1	0
1	1	1	1	1

Determine Function and fill the table implemented by the following Mux.



A	B
0	0
0	1

2 Inputs acting C and D hence making group of 4.

when Selection Input is 0 1 we and of C and D

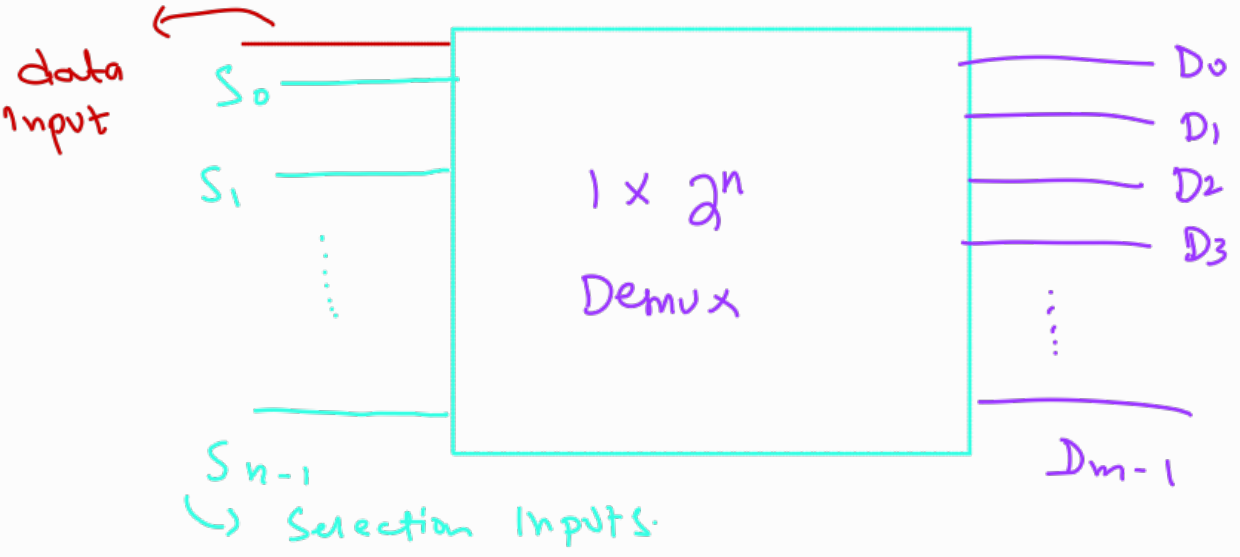
A	B	C	D	F	
0	0	0	0	1	
0	0	0	1	1	
0	0	1	0	1	
0	0	1	1	1	
0	1	0	0	0	And
0	1	0	1	0	
0	1	1	0	0	
0	1	1	1	0	
1	0	0	0	1	NOR
1	0	0	1	0	
1	0	1	0	0	
1	0	1	1	0	
1	1	0	0	0	
1	1	0	1	0	
1	1	1	0	0	
1	1	1	1	0	

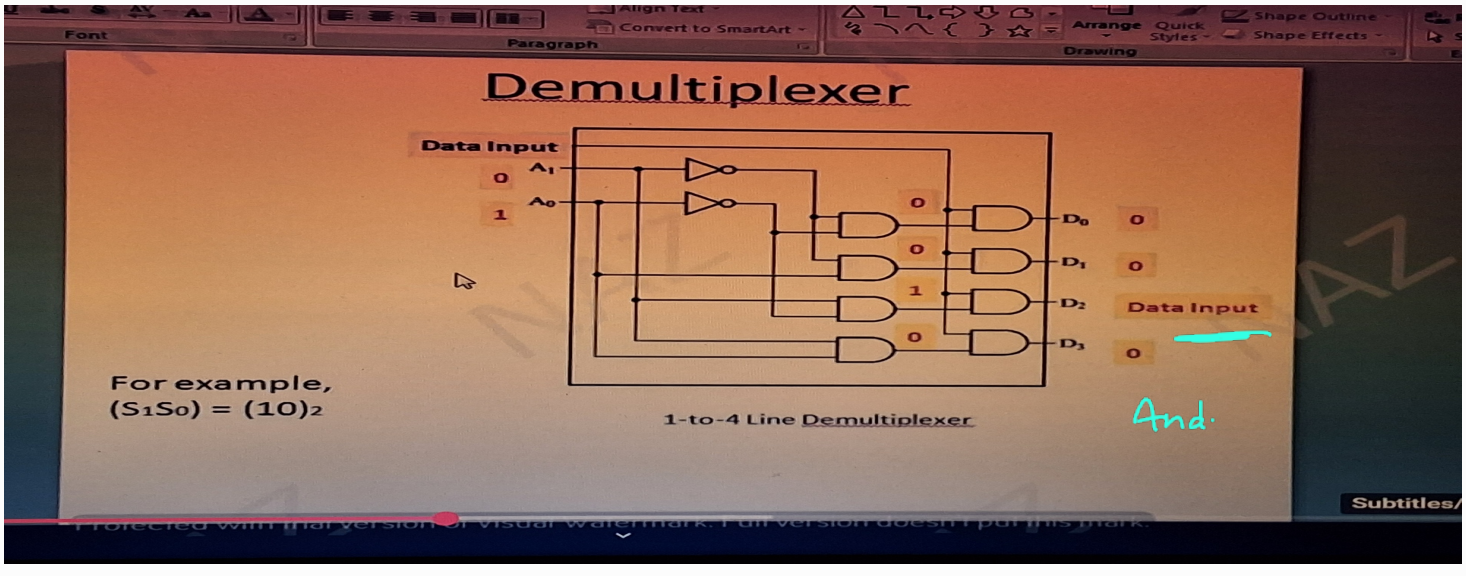
DeMux (Inverse of multiplexer)

→ Recieves information from single line | one data input.

→ Transmits 2^n output lines where n is selection inputs.

1 to m line deMux





Making Larger Mux by using Smaller Mux

Implement Boolean function using MUX

$F(A, B, C) = \Sigma(1, 3, 5, 6)$

↓

n variables

$2^2 = 4 \rightarrow$ inputs

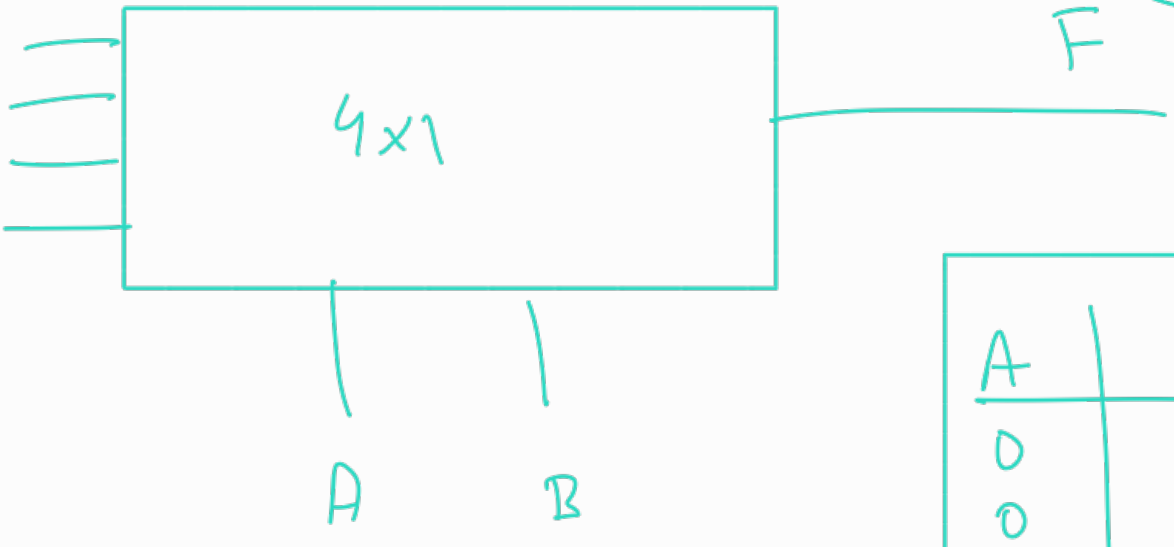
no of selection lines = $n-1$

$= 3-1 = 2 \rightarrow$ selection lines

for e.g.

$S_0 = B$

$S_1 = A$



A	B	F
0	0	I_0
0	1	I_1
1	0	I_2
1	1	I_3

Now Make T.T for $F(A,B,C) = \Sigma(1,3,5,6)$

	A	B	C	F
0	0	0	0	0
1	0	0	1	1
2	0	1	0	0
3	0	1	1	1
4	1	0	0	0
5	1	0	1	1
6	1	1	0	1
7	1	1	1	0

now how C will be consider as input

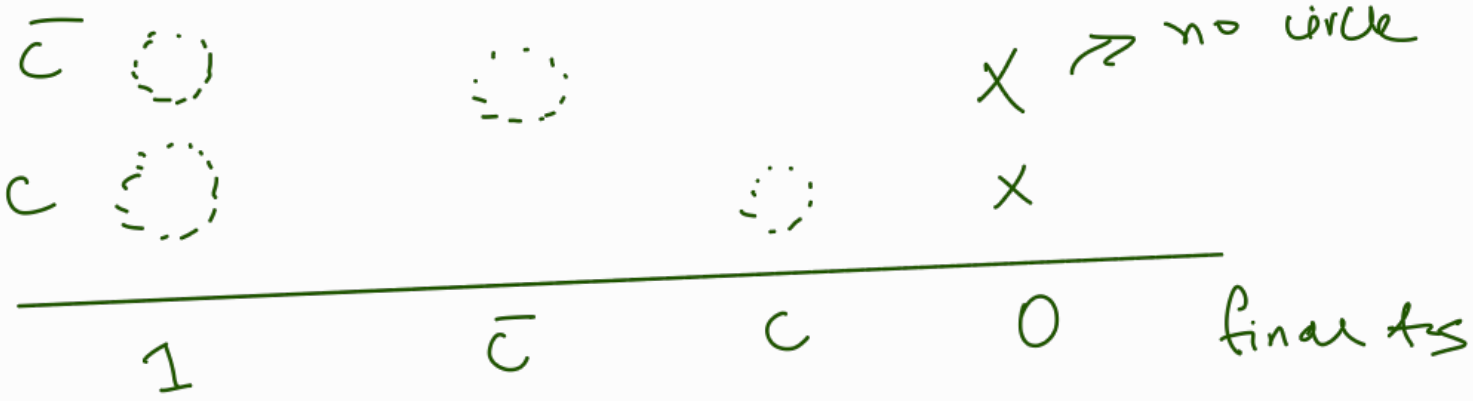
	I_0	I_1	I_2	I_3
C'	0	2	4	(6)
C	(1)	(3)	(5)	7

For Sop look where $\bar{C}$ value is 0 in t.t write decimal

↳ where C value is 1 in t.t

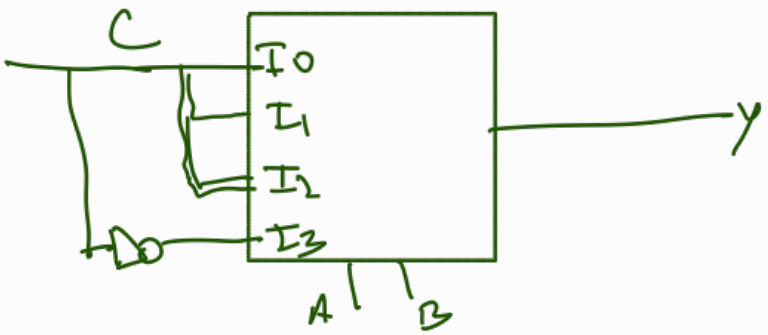
Now circle according to the given original equation.

Rules for circle



	I_0	I_1	I_2	I_3
C'	0	2	4	(6)
C	(1)	(3)	(5)	7
	C	C	C	$\bar{C}$

these are the four inputs.



Another example:-

$$F(A,B,C,D) = \Sigma(0,1,2,3,4,9,13,14,15)$$

	A	B	C	D	F
0	0	0	0	0	1
1	0	0	0	1	1
2	0	0	1	0	1
3	0	0	1	1	1
4	0	1	0	0	1
5	0	1	0	1	0
6	0	1	1	0	0
7	0	1	1	1	0
8	1	0	0	0	0
9	1	0	0	1	1
10	1	0	1	0	0
11	1	0	1	1	0
12	1	1	0	0	0
13	1	1	0	1	1
14	1	1	1	0	1
15	1	1	1	1	1

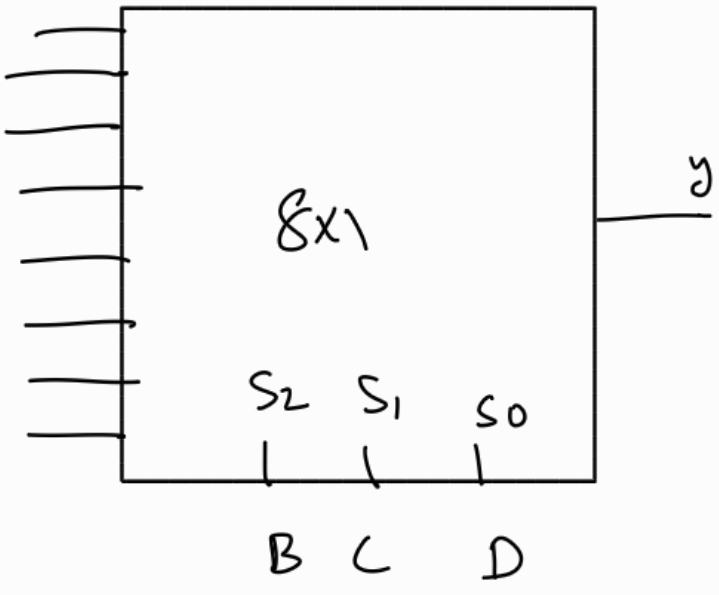
no of selection lines

$4-1 = (3)$

no of input lines

$2^3 = (8)$

using 8x1 mux

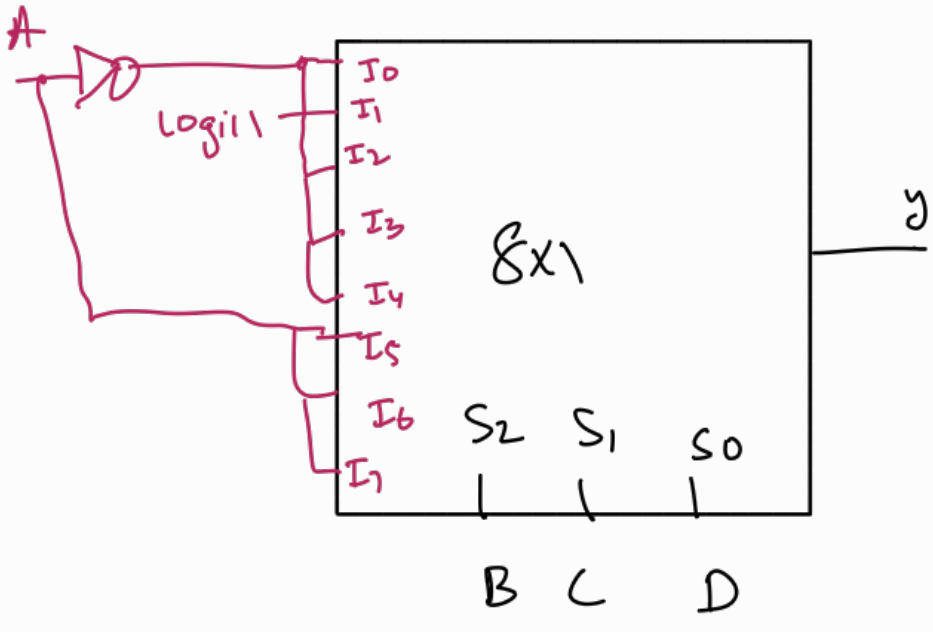


now finding A as input By looking at Truth table

	I_0	I_1	I_2	I_3	I_4	I_5	I_6	I_7
$\overline{A}$	0	1	2	3	4	5	6	7
A	8	9	10	11	12	13	14	15

Now circle $F(A,B,C,D) = \Sigma(0,1,2,3,4,9,13,14,15)$

	I_0	I_1	I_2	I_3	I_4	I_5	I_6	I_7
$\bar{A}$	0	1	2	3	4	5	6	7
A	8	9	10	11	12	13	14	15
	$\bar{A}$	1	$\bar{A}$	$\bar{A}$	$\bar{A}$	A	A	A



Implementing Boolean function using MUX and don't care

$F(A, B, C, D) = \Sigma m(0, 2, 6, 11, 12, 13)$

$\Sigma d(3, 8, 14)$

Total selection lines = $4 - 1 = 3$

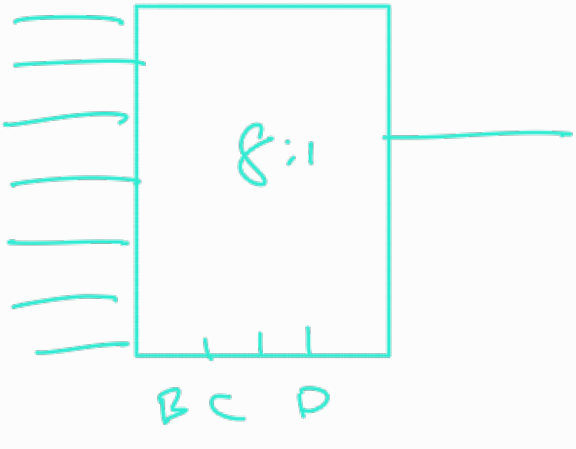
$2^3 = 8$ using **8x1 MUX**

	A	B	C	D	F
0	0	0	0	0	0
1	0	0	0	1	0
2	0	0	1	0	1
3	0	0	1	1	X
4	0	1	0	0	0
5	0	1	0	1	0
6	0	1	1	0	1
7	0	1	1	1	0
8	1	0	0	0	X
9	1	0	0	1	0
10	1	0	1	0	0
11	1	0	1	1	1
12	1	1	0	0	1
13	1	1	0	1	1
14	1	1	1	0	X
15	1	1	1	1	0

I_0 I_1 I_2 I_3 I_4 I_5 I_6 I_7

A: (8) 9 10 (11) (12) (13) (14) 15

$\bar{A}$: (0) 1 (2) (3) 4 5 (6) 7



□ → don't care.

$I_0 = 1$ $I_1 = 0$ $I_2 = \bar{A}$ $I_3 = 1$

$I_4 = 1$ $I_5 = A$ $I_6 = A$ $I_7 = 1$

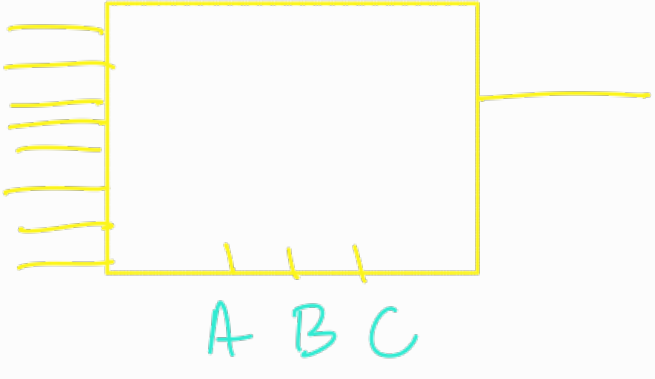
$I_8 = 0$

Implementing boolean function using Mux. POS

$f(A, B, C, D) = \pi(1, 5, 12, 15)$ By using 8:1

Mapping D

A	B	C	D	F	
0	0	0	0	1	0
0	0	0	1	0	1
0	0	1	0	1	2
0	0	1	1	1	3
0	1	0	0	1	4
0	1	0	1	0	5
0	1	1	0	1	6
0	1	1	1	1	7
1	0	0	0	1	8
1	0	0	1	1	9
1	0	1	0	1	10
1	0	1	1	1	11
1	1	0	0	0	12
1	1	0	1	1	13
1	1	1	0	1	14
1	1	1	1	0	15

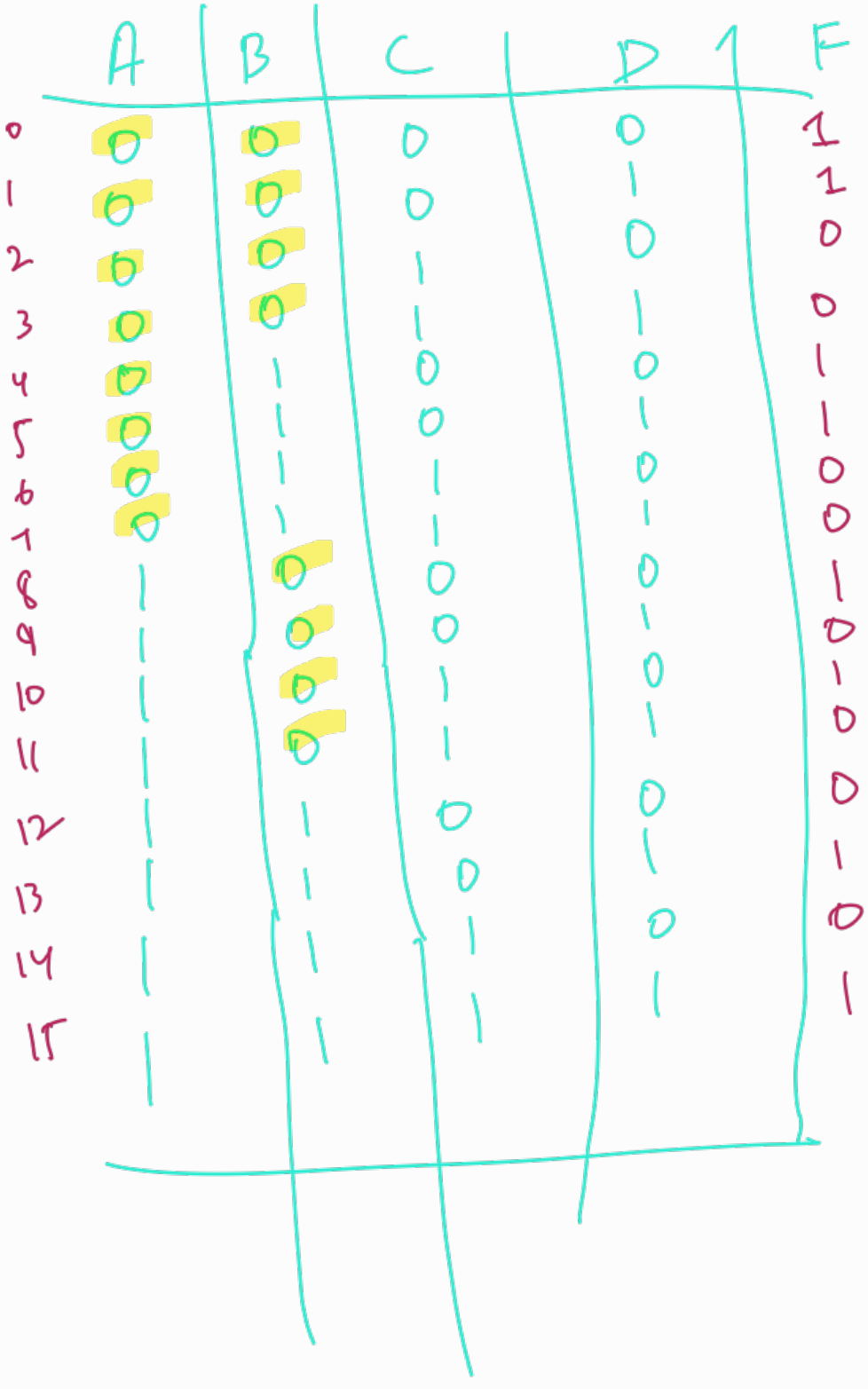


	I_0	I_1	I_2	I_3	I_4	I_5	I_6	I_7
D	0	2	4	6	8	10	12	14
$\bar{D}$	1	3	5	7	9	11	13	15
	$\bar{D}$	0	$\bar{D}$	0	0	0	D	$\bar{D}$

In pos $\bar{x}$ is used for 1 and x is used for 0

$G(A,B,C,D) = \Sigma m\,(0,1,4,5,8,10,13,15)$

2 Selection lines considering C and D



$2^n = 4$

$n = 2$

0, 1, 2, 3

A and B are inputs variables

$\bar{A}$ = 0 in TT

A = 1 in TT

	I_0	I_1	I_2	I_3
$\bar{A}\bar{B}$	0	1	2	3
$\bar{A}B$	4	5	6	7
$A\bar{B}$	8	9	10	11
AB	12	13	14	15

$\bar{A}\bar{B}$ ↓ means Both values must be 0.

Looking vertically:

$$I_0 = \bar{A}\bar{B} + \bar{A}B + A\bar{B}$$

$$= \bar{A}(\bar{B} + B) + A\bar{B}$$

$$= \bar{A} + A\bar{B}$$

$$I_1 =$$

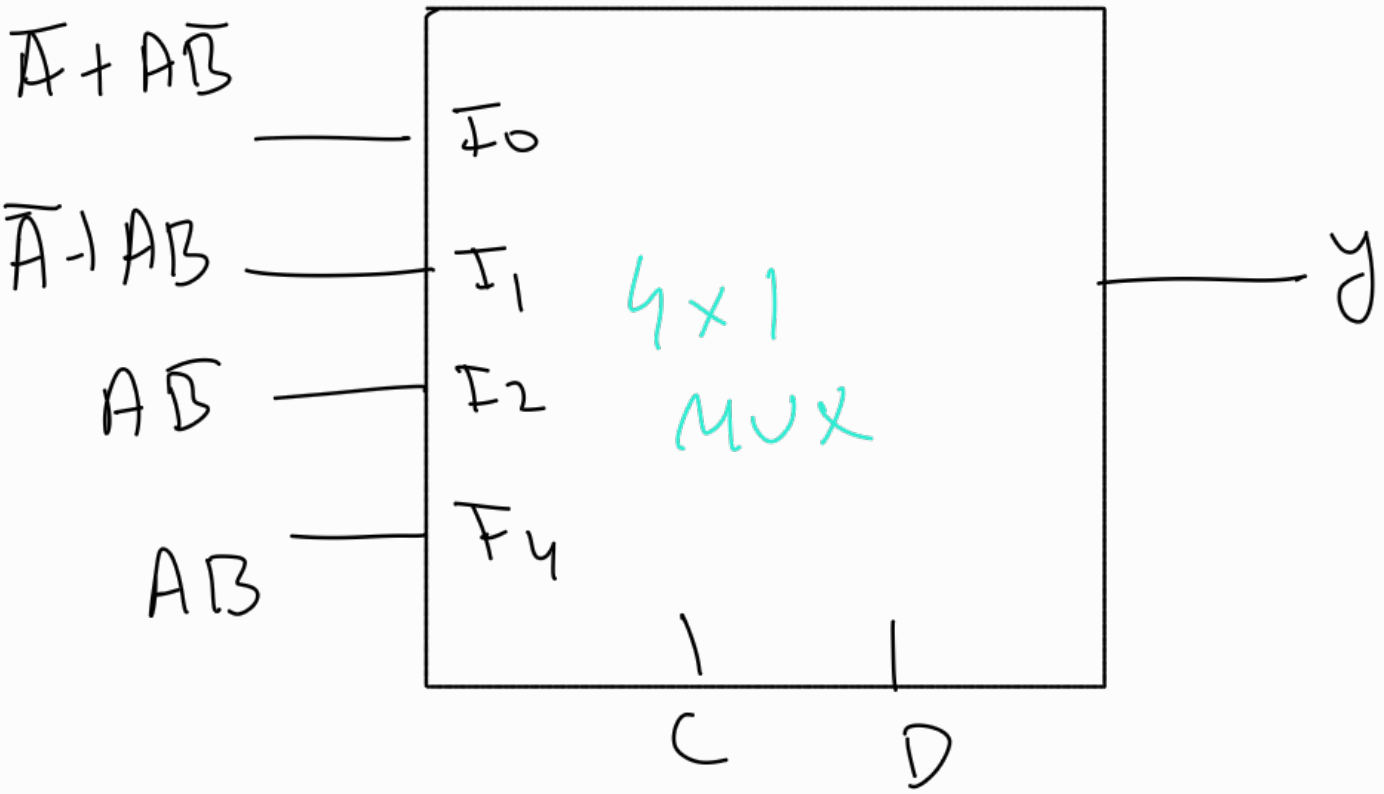
$$\bar{A}\bar{B} + \bar{A}B + AB$$

$$= \bar{A}(\bar{B} + B) + AB$$

$$= \bar{A} + AB$$

$I_2 = A\bar{B}$

$I_3 = AB$



Another way:

Implement $\Sigma m(0, 1, 2, 3, 4, 10, 11, 14, 15)$ using 8:1 MUX

$2^n = 8$ (n=3) Select lines

	A	B	C	D	F	
0	0	0	0	0	1	F=1
1	0	0	0	1	1	I_0
2	0	0	1	0	1	
3	0	0	1	1	1	I_1 F=1
4	0	1	0	0	1	I_2 $F=\bar{D}$
5	0	1	0	1	0	
6	0	1	1	0	0	I_3 F=0
7	0	1	1	1	0	
8	1	0	0	0	0	I_4 F=0
9	1	0	0	1	0	
10	1	0	1	0	1	I_5 F=1
11	1	0	1	1	1	
12	1	1	0	0	0	I_6 F=0
13	1	1	0	1	0	
14	1	1	1	0	1	I_7 F=1
15	1	1	1	1	1	

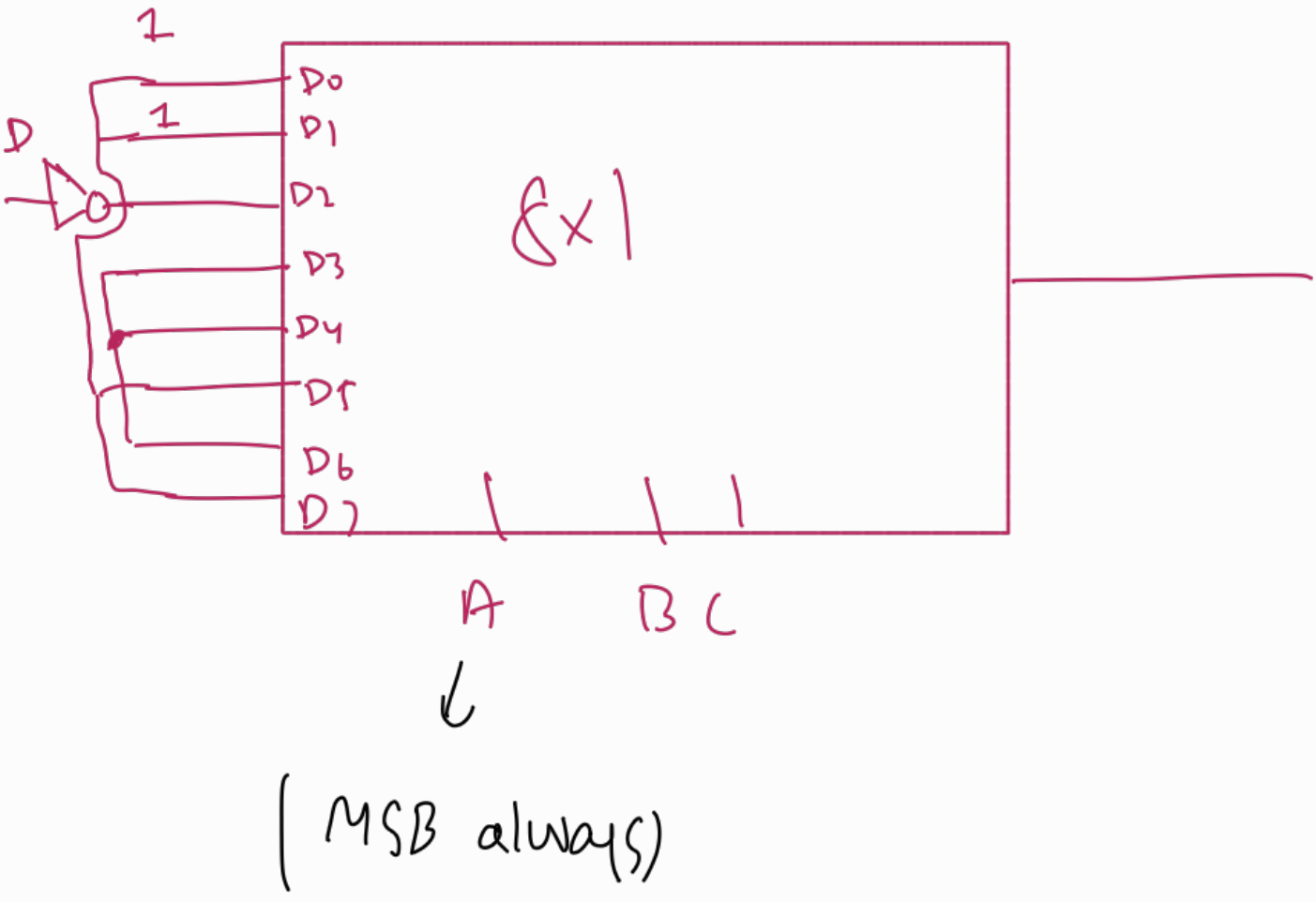
Total inputs are 8 hence dividing the 16 rows TT into 2 section

$S_2 = A$

$S_1 = B$

$S_0 = C$

(D will be the input variables.)



(MSB always)

$f(A,B,C,D) = \Sigma m(0,1,4,5,8,10,13,15)$

By using 4x1 mux.

$2^n = 4$

$n = 2$

Selection lines.

Dividing the t.t

So we have

4 inputs

Select lines are A and B

C and D are data inputs.

A	B	C	D	F	
0	0	0	0	1	$F = \bar{C}$
0	0	0	1	1	
0	0	1	0	0	
0	0	1	1	0	
0	1	0	0	1	$F = \bar{C}$
0	1	0	1	1	
0	1	1	0	0	
0	1	1	1	0	
1	0	0	0	1	$F = \bar{D}$
1	0	0	1	0	
1	0	1	0	1	
1	0	1	1	0	
1	1	0	0	0	$F = D$
1	1	0	1	1	
1	1	1	0	0	
1	1	1	1	1	

